

2025-10-09 - Valentini (3)

Previous lecture: [2025-10-03 - Valentini \(2\)](#)

Next lecture: [2025-10-10 - Valentini \(4\)](#)

Solitons

Solutions in the form of solitons waves

(similar to a gaussian wave packet)

pic01 - soliton

with

$$M = \frac{v_0}{c}, \quad \xi = x - Mt$$

and bound condition for $\xi \rightarrow \infty$ (with $()' \rightarrow \frac{d}{d\xi}$)

$$\phi' = 0, \quad \phi = 0$$

$$v' = 0, \quad v = 0$$

$$n' = 0, \quad n = 1$$

$$\begin{cases} -M \frac{\partial n}{\partial \xi} + \frac{\partial(nv)}{\partial \xi} = 0 & (A) \\ -M \frac{\partial v}{\partial \xi} + v \frac{\partial v}{\partial \xi} = -\frac{\partial \phi}{\partial \xi} & (B) \\ -\frac{\partial^2 \phi}{\partial \xi^2} = n - \exp(\phi) & (C) \end{cases}$$

From (A) (using also the boundary conditions)

$$\frac{\partial}{\partial \xi} [n(v - M)] = 0$$

$$\implies nv - nM = \xi \rightarrow \infty -M$$

$$\implies n = \frac{M}{M - v}$$

From (B)

$$\frac{\partial}{\partial \xi} \left[-Mv + \frac{v^2}{2} + \phi \right] = 0$$

$$\implies -Mv + \frac{v^2}{2} + \phi = \xi \rightarrow \infty 0$$

$$\implies v^2 - 2Mv + 2\phi = 0$$

$$\implies (M - v)^2 - M^2 + 2\phi = 0$$

$$\implies (M - v) = M^2 - 2\phi$$

We only take the positive solution, since the negative one gives imaginary solutions
 We use the result from (A)

$$\Rightarrow n = \frac{M}{M^2 - 2\phi}$$

and now, substituting in (C)
 (we use ϕ to indicate $(\partial^2 \phi \partial \xi^2)$)

$$\begin{aligned} -\phi &= n - \exp(\phi) \\ \Rightarrow -\phi &= \frac{M}{M^2 - 2\phi} - \exp(\phi) \\ \Rightarrow -\phi' \phi &= \frac{M\phi'}{M^2 - 2\phi} - \exp(\phi)\phi' \\ \Rightarrow -\phi' \phi d\xi &= \frac{M\phi'}{M^2 - 2\phi} d\xi - \exp(\phi)\phi' d\xi \end{aligned}$$

The first integral becomes

$$-\phi' \phi d\xi = -\frac{1}{2} \frac{d}{d\xi} \phi'^2 d\xi = -\frac{\phi'^2}{2}$$

The second integral becomes

$$M \frac{\phi'}{M^2 - 2\phi} d\xi = -MM^2 - 2\phi$$

And the third

$$-\exp(\phi)\phi' d\xi = -\exp(\phi)$$

so we have

$$\begin{aligned} -\frac{\phi'^2}{2} &= -MM^2 - 2\phi - \exp(\phi) + \\ \Rightarrow \frac{\phi'^2}{2} &= MM^2 - 2\phi + \exp(\phi) + \end{aligned}$$

we find using boundary condition ($\xi \rightarrow \infty$)

$$\begin{aligned} 0 &= M^2 + 1 + \\ \Rightarrow &= -(1 + M^2) \end{aligned}$$

so

$$\Rightarrow \frac{\phi'^2}{2} = MM^2 - 2\phi + \exp(\phi) - (1 + M^2) \quad (\text{e01})$$

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Now we need some assumptions:

$$\phi \ll 1, \quad 0 \ll M \ll 1$$

with

$$M = M - 1 = \frac{v_0}{c_s} - 1$$

the second condition tells us

$$0 < M \implies v_0 < c_s$$

$$M > 1 \implies v_0 > c_s$$

Also, later we will see that these two conditions are the same condition

So

$$M = 1 + M, \quad M^2 = 1 + 2M + M^2$$

and

$$M^2 - 2\phi = \left(1 + 2M + \cancel{M^2} - 2\phi\right)^{12} \left[1 - 2(M - \phi)\right]^{12}$$

and we can Taylor expand

$$M^2 - 2\phi = 1 + \frac{2}{2} - \frac{2}{2} + \frac{1}{1} -$$

and

$$\exp(\phi) = 1 + \phi + \frac{\phi^2}{2} + \frac{\phi}{2} +$$

and

$$1 + M^2 = 1 + 1 + 2M + \cancel{M^2} = 2 + 2M$$

Substituting into the [eq. 01](#), multiplying and only keeping the terms with the following powers

$$M, \phi, \phi^2, \phi, M\phi, M\phi^2$$

we obtain

$$\begin{aligned} \frac{\phi'^2}{2} &= MM^2 - 2\phi + \exp(\phi) - (1 + M^2) \quad (\text{e 1}) \\ \frac{\phi'^2}{2} &= (1 + M) \cdot \left[1 + (M - \phi) - \frac{1}{2}(M - \phi)^2 + \frac{1}{2}(M - \phi)\right] + \\ &\quad + \left(1 + \phi + \frac{\phi^2}{2} + \frac{\phi}{2}\right) + (2 + 2M) \\ &\quad - \frac{\phi}{2} + \frac{1}{2}M\phi^2 - \frac{M\phi^2}{2} + \frac{\phi}{2} \\ &\quad - \frac{\phi}{2} + M\phi^2 \end{aligned}$$

which is non linear yet.

We rewrite it as

$$\phi'^2 = 2M\phi^2 \left[1 - \frac{\phi}{M} \right]$$

$$\Rightarrow \phi' = 2M\phi \left[1 - \frac{\phi}{M} \right]^{1/2}$$

where we take only the positive value of the square root.

Now we solve using separation of variables

$$\frac{1}{\phi} \frac{d\phi}{1 - \frac{\phi}{M}} = 2M d\xi$$

and

$$\frac{1}{\phi} \frac{d\phi}{1 - \frac{\phi}{M}} = 2M d\xi = \xi 2M + 1$$

On the left side we apply two change of variables

$$1. \quad \phi = \xi M, \text{ so } d\phi = M d\xi$$

$$2. \quad \eta = 1 - \frac{\phi}{M}, \text{ so } d\eta = -\frac{1}{M} d\phi, \quad d\phi = -M d\eta$$

$$\begin{aligned} \frac{1}{\phi} \frac{d\phi}{1 - \frac{\phi}{M}} &= - \frac{1}{\xi M} \frac{M d\xi}{\eta} = \\ &= -2 \frac{d\xi}{1 - \eta} = \\ &= -2 \left(\frac{1}{\eta} \right) + 2 = \\ &= -2 \left[\left(1 - \frac{\phi}{M} \right)^{-1} \right] + 2 \end{aligned}$$

so we have

$$\begin{aligned} -2 \left[\left(1 - \frac{\phi}{M} \right)^{-1} \right] + 2 &= \xi 2M + 1 \\ \Rightarrow -2 \left[\left(1 - \frac{\phi}{M} \right)^{-1} \right] &= \xi \frac{2M}{2} + \frac{1+2}{2} \\ \Rightarrow -1 - \frac{\phi}{M} &= \left[\xi \frac{M}{2} + \right] \\ \Rightarrow \phi &= M \left(1 - \left[\xi \frac{M}{2} + \right] \right) \\ \Rightarrow \phi &= M e^{-2} \left(\xi \frac{M}{2} + \right) \end{aligned}$$

For every $\phi \rightarrow 0$

So we choose $\xi = 0$ without any loss of generality and

$$\phi = M e^2 \left(\frac{M}{2} \xi \right) = M e^2 \left[\frac{M}{2} (x - Mt) \right]$$

The maximum amplitude of ϕ is M

That's why

$$\phi \approx 1 \quad M \approx 1$$

Solitons are stable waves and are more easily detectable than others

Space probes captures them

IRS -> Isolated Electrostatic Structures

they are everywhere in the magnetosphere

Kinetik theory of plasma

Vlasov equation

$$\frac{\partial \alpha}{\partial t} + \bar{v} \nabla_{\alpha} + \frac{\alpha}{m_{\alpha}} \left[\bar{E} + \frac{\bar{v} \bar{B}}{c} \right] \cdot \nabla_{\bar{\alpha}} = 0$$

There are different methods to obtain it

- from Liouville equation
- using the Klimontovich derivation (which we will study)

(we will replace the pedix α with s , in order to follow the notation of the book)

(m is an index like s , which we will use to do a trick later)

Outline of the Klimontovich method:

1. We start from an ensemble of particles
2. We study the equations of motion

$$\begin{cases} \bar{x}_s(t) = \bar{v}_s(t) \\ \bar{v}_s(t) = \frac{s}{m_s} \left[\bar{E}_m(\bar{x}, t) + \frac{\bar{v} \bar{B}(\bar{x}, t)}{c} \right] \end{cases}$$

We will use the following math instruments:

- Dirac Delta
- Redisuals Theorem
- Laplace Transform
- Green's Function

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